

Computer Simulation

Lesson 2: Introduction to MATLAB

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Special thanks to: Dr. Pham Thi Huong

Lesson Organization

- Getting Started (Start and Quit MATLAB)
- Entering Commands (Variable)
- Array Expressions (Arrays, Scalars, Vectors, Matrices)
- Matrix Operations
- Matrix Practice I (Operations)
- Matrix Practice II (Application)
- Matrix Practice III (Calculation)
- Home Work

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Getting Started

- MATLAB environment
 - Command Window: main window
 - Enter variables
 - Run functions, m-file scripts
 - Command History: Log previously run statements
 - Workspace: List all variables
 - Current Directory: List all m files in the selected directory
- The “>>”: prompt
- Quit
 - Type “quit” at the prompt
 - Menu → File → Exit

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Getting Started

Useful Commands

Commands	Description
info	Display contact information for the product
demo	Demos of several options in MATLAB
help	Giving a more detail description
help help	Explain how help command works
helpbrowser	Open a Help window
lookfor	Showing the associated commands for a searching word
clc	Clean the display window, the value of variables are remained
Esc	Delete current typing line
Ctrl-C	Quit the current program

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Introduction to MATLAB

- Calculator: solving numerical problems
- Programming language (.m file)
- Doing task on 2 ways:
 - Directly on command window
 - Save program to a file and execute program by using the saved file

Example

- Command window: `>>1+2`
- Save to a file: ~~Example1+2.m~~
Example1_2.m
- Note:
 - MATLAB distinguish between round brackets () and square brackets []: `a()` # `a[]`
 - In all situations, MATLAB always distinguish between uppercase letter and lowercase letters: `a`#`A`

Display Result: Comma, Semi-Colon

- Show both results
`x=1, y=2`
- Show one results
`x=1;y=2*x`
`x=1,y=2*x;`
- Hidden all results
`x=1;y=2*x;`
- Echo: display statements during function execution

Entering Commands

- Variable is created with an assignment statement:
`variable_name = expression`
- Example: `x = 5`
- Note:
 - Assignment operator “=” does not mean equality “==”
 - Variable name must always be on the left: E.g., `5 = x` ???
 - Name:
 - Must begin with a letter (e.g., `My_1`)
 - Cannot have a space
 - Difference between upper and lowercase letters (e.g., `My` # `MY` # `my`)
 - Cannot MATLAB keywords (check: `iskeyword`)

Variable

Useful Commands

Commands	Description
who	Shows all of the variables in Matlab workspace
whos	Shows all information of the variables (name, matrix size)
clear	Clear all variables in the Workspace
clear x	Clear a particular variable x
what	List all m files in the Working Folder

Array Expressions

- Array: any set of numbers arranged in a rectangular pattern
 - Example: two dimension, three dimension
- Matrix, vector, scalar
 - Matrix: 2D array
 - $m \times n$: m rows, n columns
 - Vector: 1D array
 - $1 \times n$: row vector
 - $n \times 1$: column vector
 - Scalar: 1×1 matrix (a single number)

Create an array

- Scalar: $x = 1$
- Vector
 - Row vector: $R = [1,2,3]$ or $R = [1 \ 2 \ 3]$
 - Column vector: $C = [1;2;3]$
- Matrix: $M = [1,2,3;4,5,6]$ or $M = \begin{bmatrix} 1,2,3 \\ 4,5,6 \end{bmatrix}$
- Note:
 - Elements of one row are separated by **space** or the **commas “,”**
 - Rows are separated by hitting **enter key** or the **semicolon “;”**

Special Matrices

- **zeros(m,n)**: $m \times n$ matrix whose element are zero
 - **zeros(n)**: $n \times n$ matrix of zeros
 - **zeros(A)**: $m \times n$ matrix of zero, where m and n are based on the size of the existing matrix A
- **ones(m,n)**: $m \times n$ matrix whose element are 1
 - **ones(n)**: $n \times n$ matrix whose element are 1
 - **ones(A)**: $m \times n$ matrix of 1, where m and n are based on the size of the existing matrix A
- **eye(n)**: $n \times n$ identity matrix with 1 on the diagonal
- **diag(X)**: Diagonal matrix with the value of X on the diagonal ($X = [1,3,5,7]$, **diag(X)**)
- **M=magic(n)**: $\text{sum}(M) = \text{sum}(M')' = \text{sum}(\text{diag}(M)) = \text{sum}(1:n^2)/n$
- **rand(m,n)**: $m \times n$ matrix of random number
 - **rand(A)**: matrix of random number with the size of A

Numeric expressions

Common Operators

- Scalar operators

Operators	Description
+	Addition
−	subtraction
*	Multiplication (Example: 3*3 is 9)
/	Right division (divided by, example: 9/3 is 3)
¥ \	Left division (divided into, example: 3¥9 is 3)
^	Exponentiation (Example: 3^3 is 27)
:	Colon
'	Transpose of matrix

Matrix Operations

- Matrix operators for multiplication, division, and power have a corresponding array operator that operates element-wise

Commands	Description
A.*B	Multiplies each element of A by respective element of B
A./B	Divides each element of A by the respective element of B
A¥B	Divides each element of B by the respective element of A
A.^B	Raise each element of A by the respective element of B

Examples

- A=[1 2 3; 4 5 6]
- A+5
- sin(A)
- A*inv(A): matrix multiplication computes the inner products between rows and columns
- A.*A: element-wise multiplication computes the products between elements and elements

```
>> x=[1,2;1,2]
```

```
>> x^2
```

```
>> x.^2
```

Referring to matrix elements

- Referring to matrix elements

```
A=[1 2 3; 4 5 6]
```

```
>> A(2,3) % element in the 2nd row, 3rd column
```

```
>> A(1:2,2:3) % 1st and 2nd rows, 2nd and 3rd columns
```

```
>> A(1,:) % all columns of the 1st row
```

```
>> A(:,2) % all rows of the 2nd column
```

```
>> A(:) % all elements regarded as a single column
```

Modifying matrix elements

- Add an element to a matrix
`>> A(3,5)=9`
- Delete an element from a matrix
 - Use an empty matrix `[]``>> A(1,:)=[] % delete the 1st row`
`>> A(:,2)=[] % delete the 2nd column`
- Concatenation: `B = [A,A]` or `[A A]` or `[A;A]`

Array Functions

- **length(A)**: returns the largest dimension in A
- **size(A)**: returns the number of rows and columns of A [m n]
- **max(A)**:
 - Return the largest element of a vector
 - Return a vector contains the largest element of each column
- **min(A)**: smallest number
- **sum(A)**: sum of elements in each column
- **sort(A)**: arrange each column increasingly
- `[x,k]=max(A)`: similar with `max(A)` command, values are saved in vector x, indexes are saved in vector k
- **find** function:
 - `[row,col,value]=find(A)`: find elements #0
 - `[row,col] = find(A>0 & A<10, 3)` → the first three elements >0 and <10
- Transposition: `A'(m,n) = A(n,m)`

Matrix Practice I

Given the array `A = [1 5 3 7 ; 2 6 4 8 ; 4 3 2 1]`, explain the results of the following commands:

1. <code>A'</code>	8. <code>[A(end,:)]</code>
2. <code>A(:, [1 4])</code>	9. <code>A(1:3,:)</code>
3. <code>A([2 3],[3 1])</code>	10. <code>[A; A(1:2,:)]</code>
4. <code>reshape(A,2,6)</code>	11. <code>sum(A)</code>
5. <code>A(:)</code>	12. <code>sum(A')</code>
6. <code>flipud(A)</code>	13. <code>sum(A,2)</code>
7. <code>fliplr(A)</code>	14. <code>[[A;sum(A)] [sum(A,2);sum(A(:))]]</code>

Matrix Practice II: Cryptography (encode and decode)

- Sending the message 'Welcome to SIT'
- Encoding matrix `E=[-3 -3 -4;0 1 1; 4 3 4]`
- Assign a number for each letter according to ASCII: `M= double(message)`
- Encoding and send the encoded message:
`EM=E*M` (**M should be 3xn matrix!**)
- Receive the encoded message, decode by decoding matrix `D=inv(E)` → `Z=D*EM`
- Display the message

Matrix Practice III

- Calculate the sum

$$S = 1^2 + 2^2 + \cdots + 100^2 = \sum_{i=1}^{100} i^2$$

- Calculate the product

$$P = \left(1 + \frac{1}{1}\right) \left(1 + \frac{1}{2}\right) \cdots \left(1 + \frac{1}{100}\right) = \prod_{i=1}^{100} \left(1 + \frac{1}{i}\right)$$

- Hint
 - Solution 1: using build-in function (sum/prod)
 - Solution 2: using “for” loop

Home Work

Given the array $A = [1 \ 5 \ 3 \ 7 ; 2 \ 6 \ 4 \ 8 ; 4 \ 3 \ 2 \ 1]$, provide the command that will

1. Assign the even-numbered columns of A to an array called B
2. Assign the odd-numbered rows to an array called C
3. Convert A into a 4-by-3 array
4. Compute the reciprocal of each element of A
5. Compute the square-root of each element of A

Thank you very much for your attention.